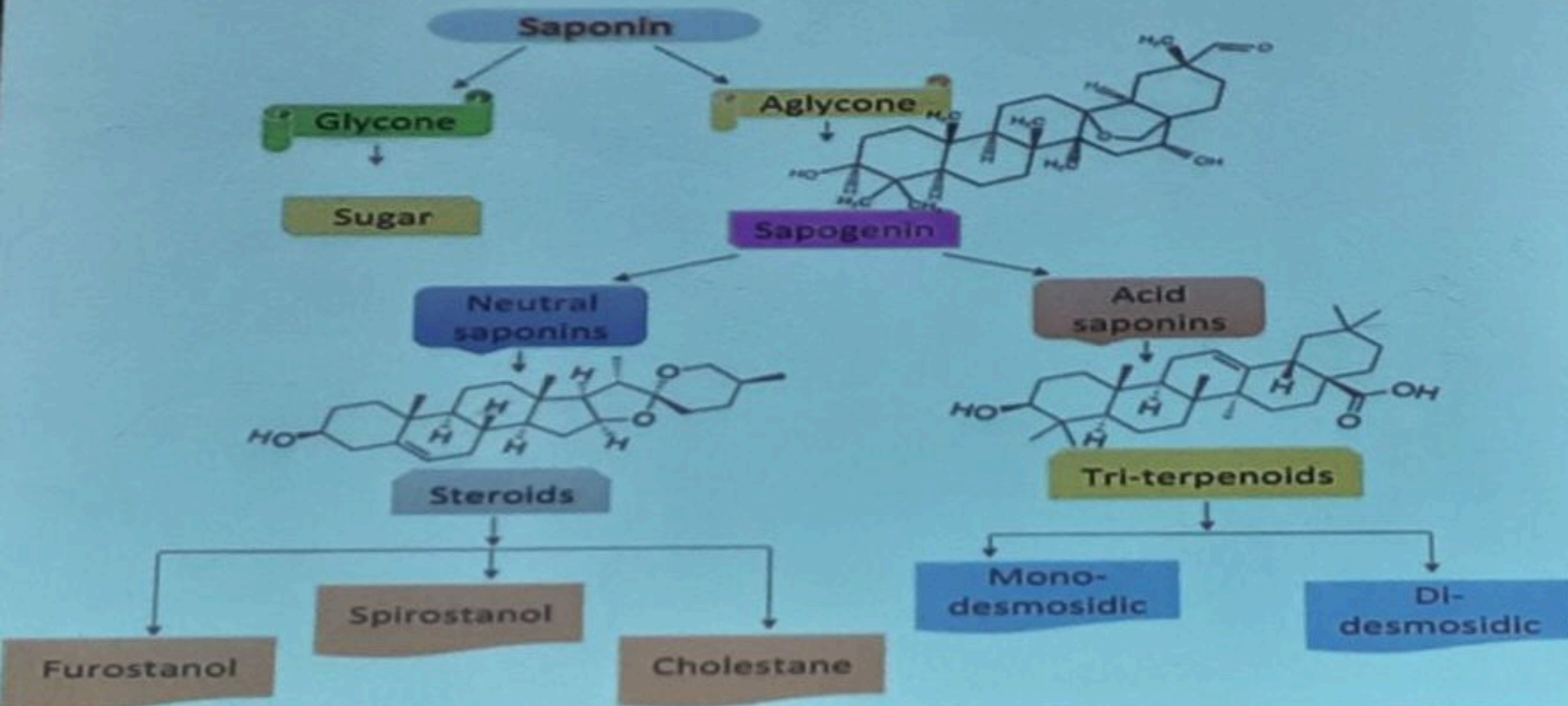


2. SAPONIN GLYCOSIDES



- Derived from the Latin word "**sapo**" due to their **surfactant** properties
 - Has **amphiphilic nature** of saponins that is **hydrophobic aglycone** backbone (sapogenin) linked to the **hydrophilic sugar** chain/s
- have long been used in many parts of the world for their detergent properties

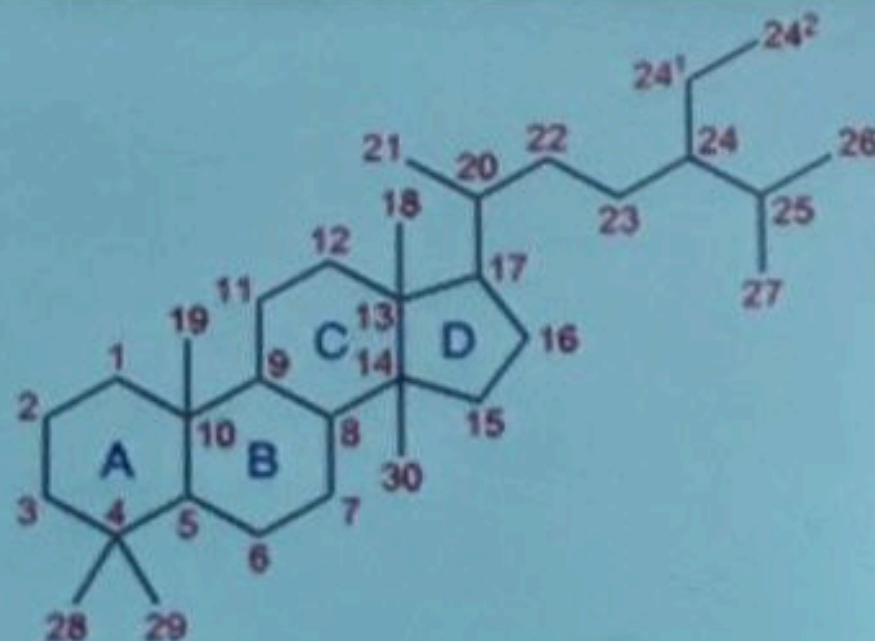
E.g. In South America bark of *Quillia saponaria*

In Europe root of *saponaria officinalis*

- Foam in aqueous solution by forming colloidal solution via **micelles** formation, which **decreases surface tension**
- Causes hemolysis of red blood cells hence toxic

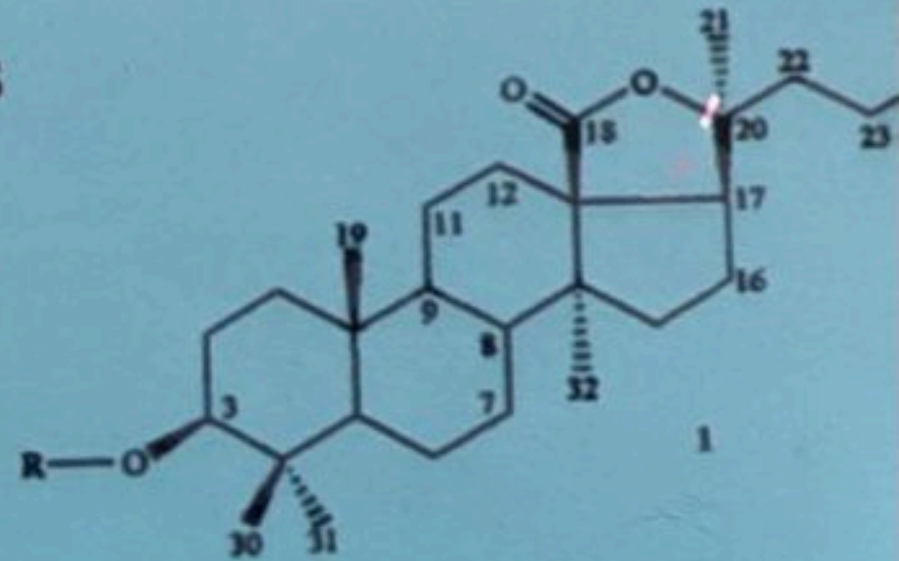
2.1. Steroidal Saponins

- Less widely distributed in nature
- Common in:
 - **Dioscoraceae** (e.g. *discoria* spp),
 - **Amaryllidaceae** (e.g. *Agavae* spp)
 - **Liliaceae** (e.g. *yucca* and *trillium* spp)
- useful starting materials for the synthesis of steroidal drugs
- Has a core structure of **cyclopentanoperhydrophenanthrene**
 - **Eg.** Diosgenin, Hecogenin, Botogenin



2. 2. Triterpenoidal Saponins

- Abundant in many dicotyledonous families:
 - Caryophyllaceae
 - Sapindaceae
 - Polygalaceae
 - Sapotaceae
- **Monodesmosides** single chain of sugars and **Bidesmosides** in which there are two sets of sugars.



General uses

- Foaming beer (spirostane-type saponins from *yucca schidigera*)
- Cleaning industrial equipment and fine fabrics
- Powerful emulsifier of certain resin, fats and fixed oils

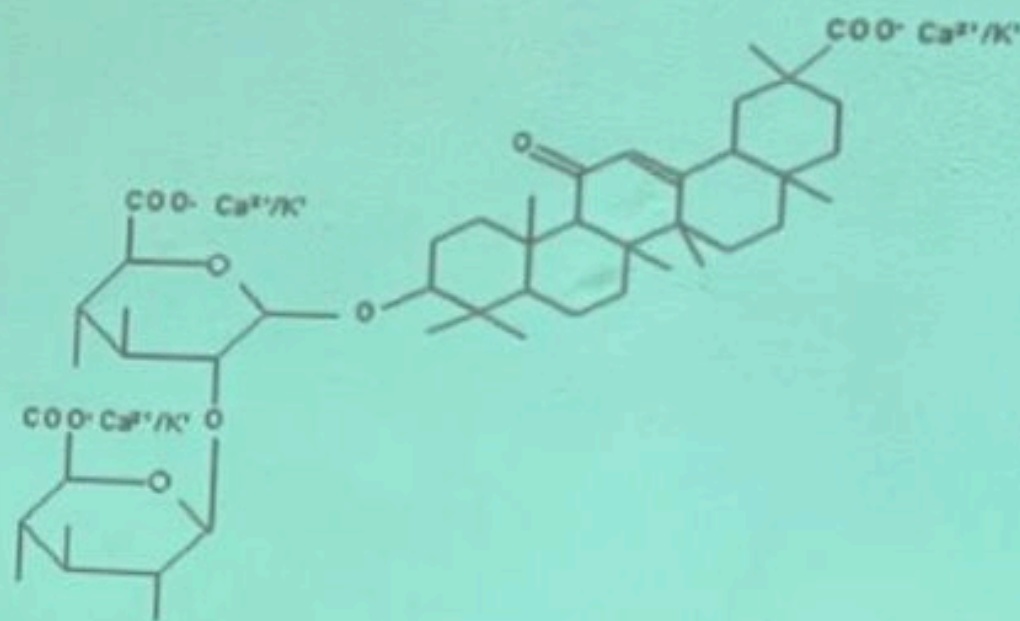
Biological uses

- Antifungal and antibacterial agents
- Lowering of blood cholesterol
- Inhibition of cancer cell growth
- precursors of partial synthesis of steroid hormones such as cortisone, progesterone, and pregnenolone
- **Expectorant, anti-inflammatory, cytotoxic, homeostatic boosters**

**Some crude drugs containing
saponins**

Licorice *Glycyrrhiza glabra*

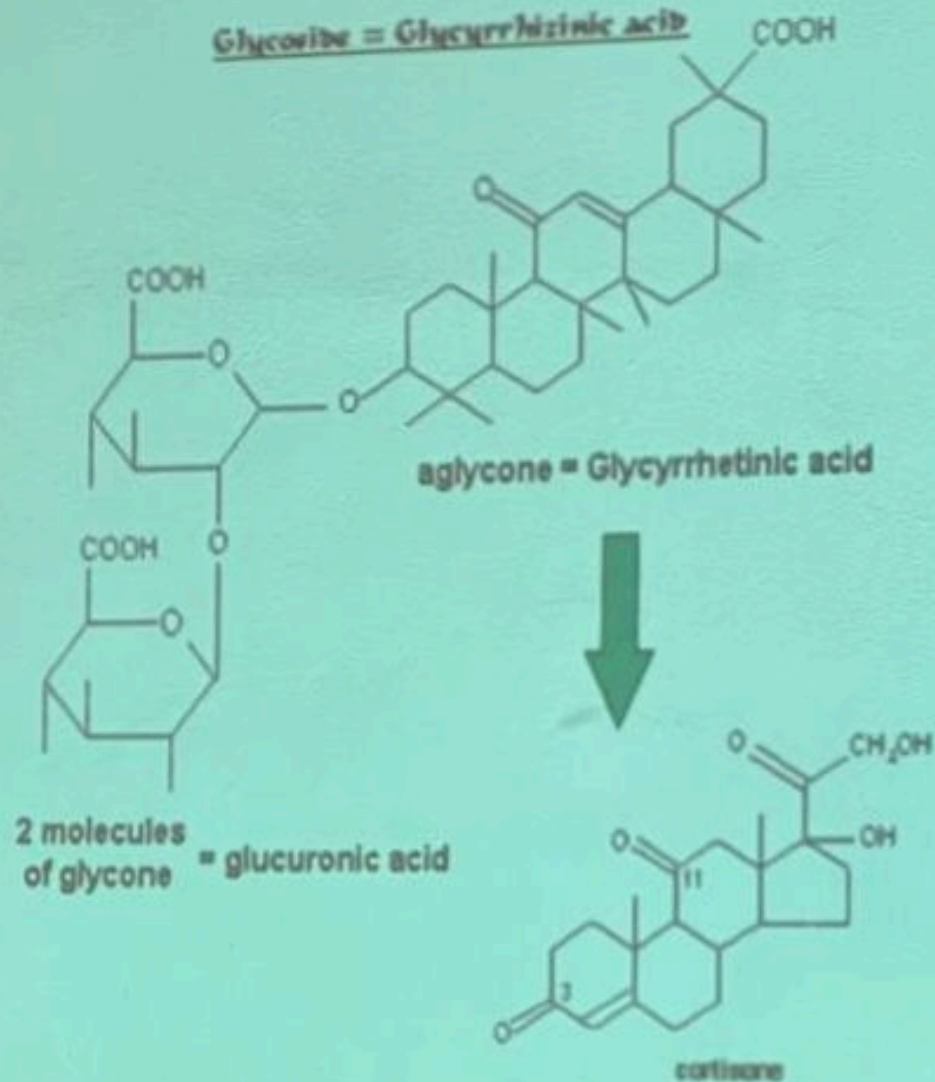
- Glycyrrhiza is the dried rhizome and roots of *Glycyrrhiza glabra*
- Contains a saponin glycoside called **Glycyrrhizin**



Glycyrrhizin = Ca^{2+} & K^{+} salts of Glycyrrhizic acid

- inhibits liver cell injury
- It inhibits the growth of several DNA and RNA viruses

Glycoside = Glycyrrhizinic acid



- **50x sweeter** than sucrose

- **expectorant and antitussive** properties

- **inhibits** the enzymes that metabolise the prostaglandins, **PGE2** and **PGF2alpha**

- Potential for the treatment of **peptic ulcer**

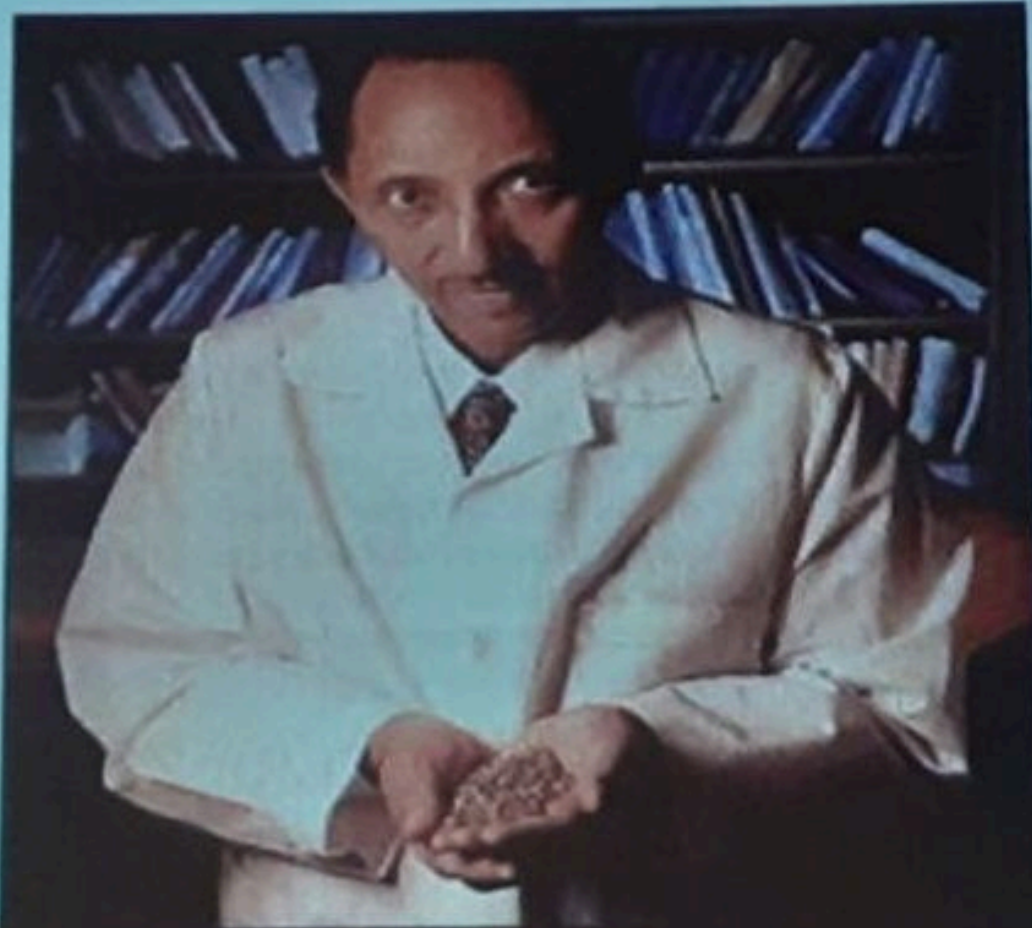
- Anti-inflammatory

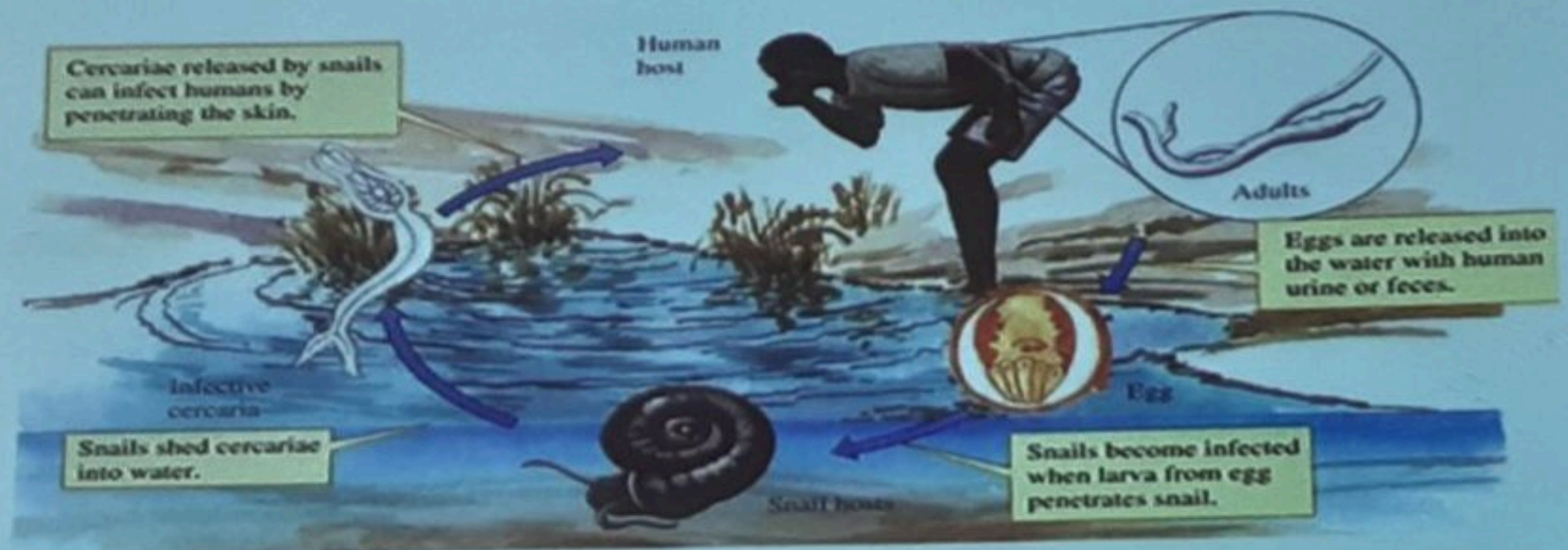
Contraindication

- Licorice should not be taken during pregnancy
- Licorice should not be given to patients with a known history of high blood pressure

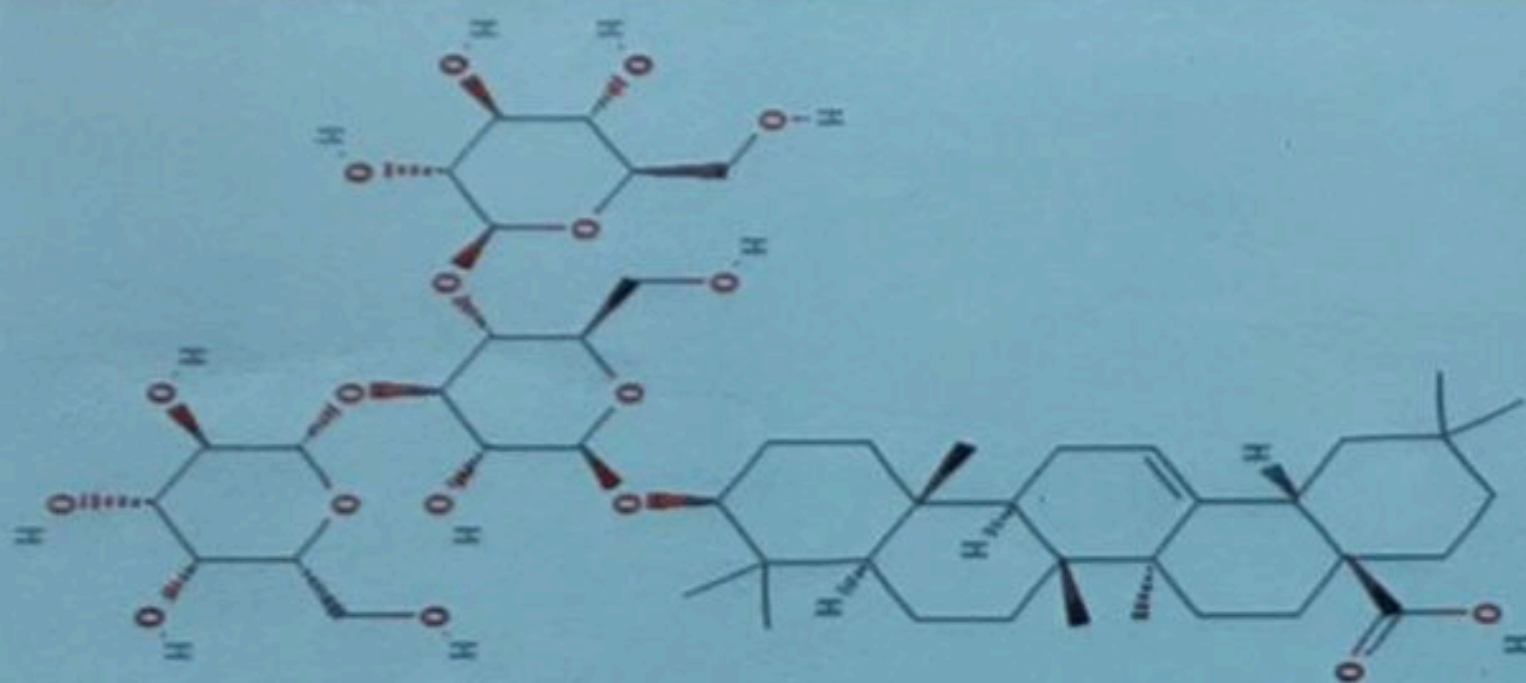


***Phytolacca dodecandra* (አንዶድ)**



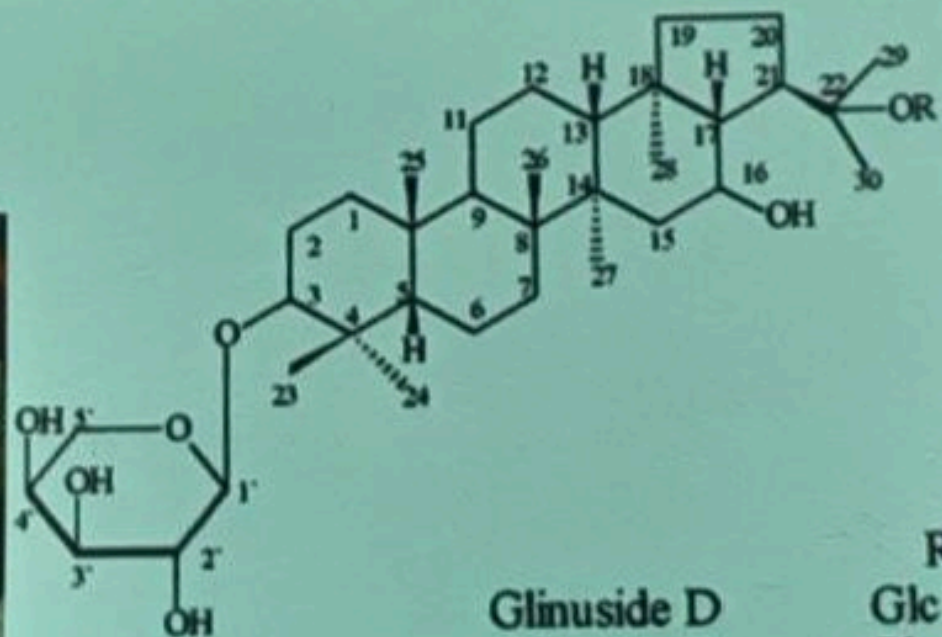


- (አንጾድ) had properties of a **molluscicide** and a **detergent**.
- It kills snails, which vectors of **schistosomiasis** ("snail fever disease"), otherwise known as **bilharzia**.



- **Lemmatoxin** isolated from *Phytolacca dodecandra*

Glinus lotoides (ਸ਼ਮ-ਫਲ)



Glinuside D
Glinuside E

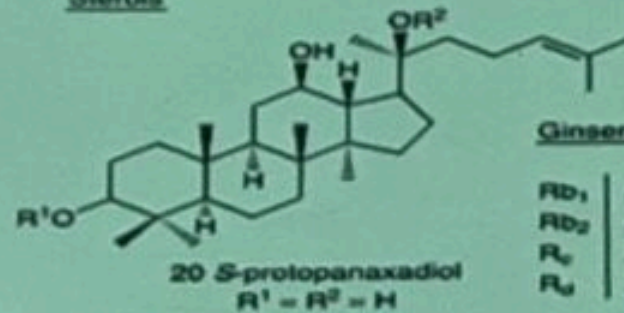
R
Glc + Rha
Glc + Glc

Anthelminthic

Ginseng

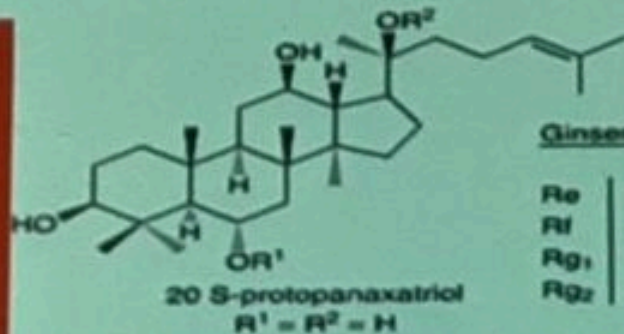


Sterols



Ginsenoside

	R^1	R^2
Rb ₁	β -D-Glc $\xrightarrow{1-2}$ β -D-Glc	β -D-Glc $\xrightarrow{1-6}$ β -D-Glc
Rb ₂	β -D-Glc $\xrightarrow{1-2}$ β -D-Glc	α -L-Ara $\xrightarrow{1-6}$ β -D-Glc
R _c	β -D-Glc $\xrightarrow{1-2}$ β -D-Glc	α -L-Arf $\xrightarrow{1-6}$ β -D-Glc
R _d	β -D-Glc $\xrightarrow{1-2}$ β -D-Glc	β -D-Glc

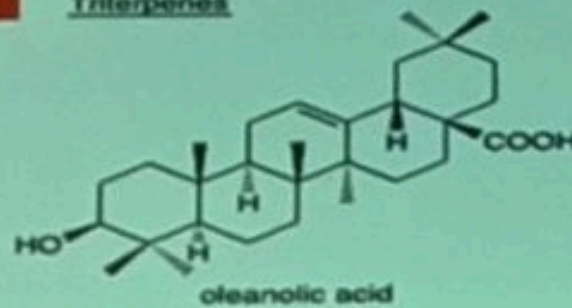


Ginsenoside

	R^1	R^2
Re	α -L-Rha $\xrightarrow{1-2}$ β -D-Glc	β -D-Glc
Rf	β -D-Glc $\xrightarrow{1-2}$ β -D-Glc	H
Rg ₁	β -D-Glc	β -D-Glc
Rg ₂	α -L-Rha $\xrightarrow{1-2}$ β -D-Glc	H

Glc = glucose, Ara = arabinose, Arf = arabinofuranoside, Rha = rhamnose

Triterpenes



Test for saponins

Foam test: 1ml solution of extract was diluted with distilled water to 20 ml and shaken in a graduated cylinder for 15 minutes. Development of **stable foam** suggests the presence of saponins.

lead acetate test : 1ml extract was treated with 1% lead acetate solution. Formation of **white precipitates** indicates the presence of saponins.

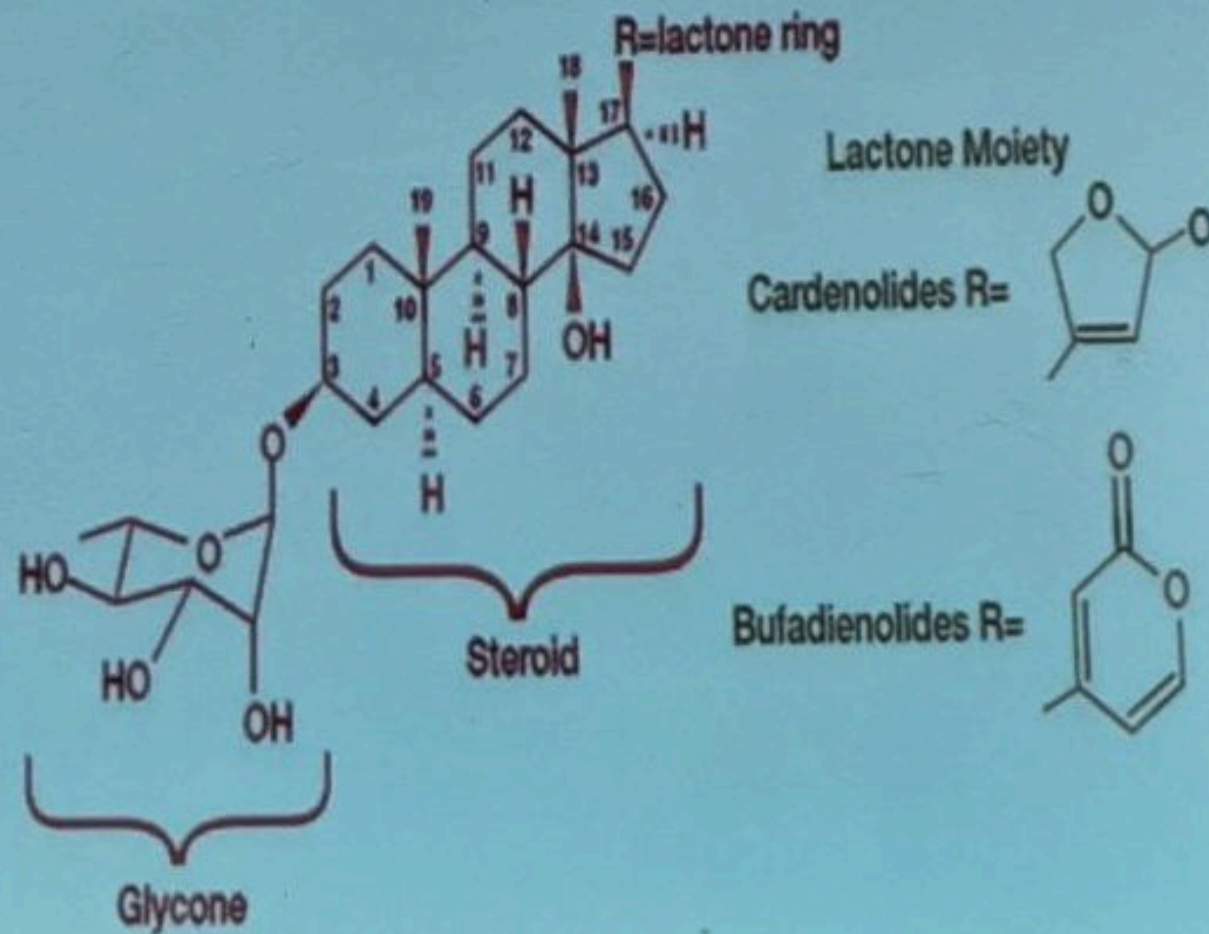
3. CARDIAC GLYCOSIDES

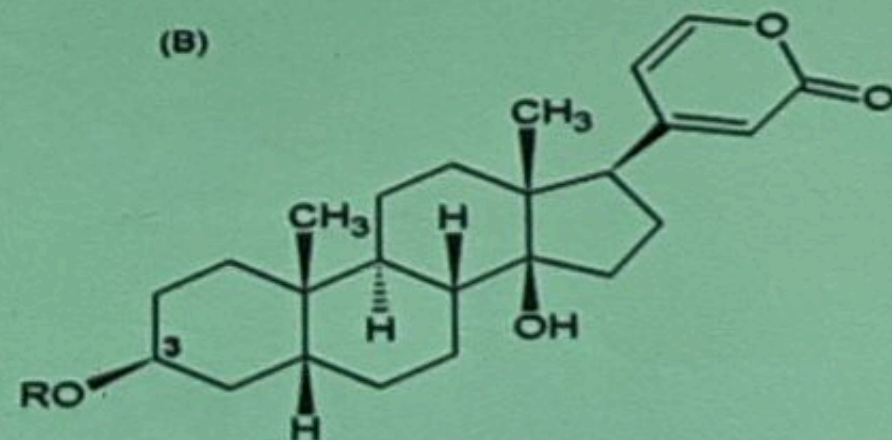
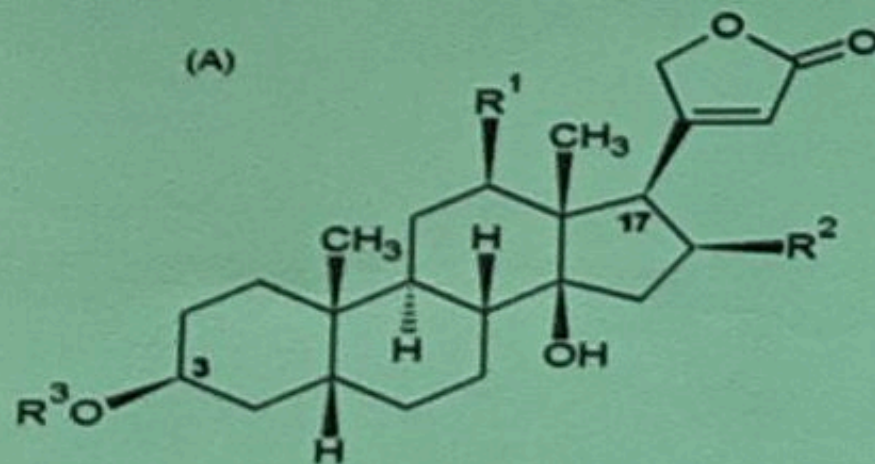
- Glycosides in which **small doses can affect the heart muscles** in a specific way
- Historical use:
 - to assassinate people, arrow poisons
- Historical sources:
 - South American toad skins, African plant extracts
- Modern use:
 - Treat congestive heart failure (dropsy)



Chemical structure

- The **R group at the 17-position** defines the class of cardiac glycoside
- plant sources provide a **5-membered unsaturated lactone**
- Animal sources give a **6-membered unsaturated lactone**





CARDENOLIDES (A)

aglycones (genins)

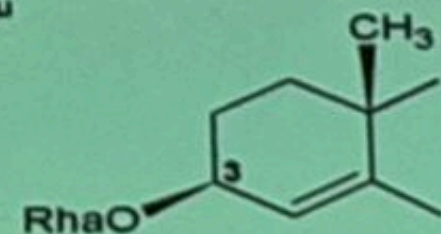
glycosides (R^3 =sugar chain)

	$R^3 = H$	$R^3 = -Dt-Dt-Dt$	$R^3 = -Dt-Dt-Dt-Glu$	$R^3 = -Dt-Dt-AcDt-Glu$
digitoxigenin	$R^1 = R^2 = H$	digitoxin	purpureaglycoside A	lanatoside A
gitoxigenin	$R^1 = H, R^2 = OH$	gitoxin	purpureaglycoside B	lanatoside B
digoxigenin	$R^1 = OH, R^2 = H$	digoxin		lanatoside C

Dt - digitoxose, Glu - glucose, Rha - rhamnose

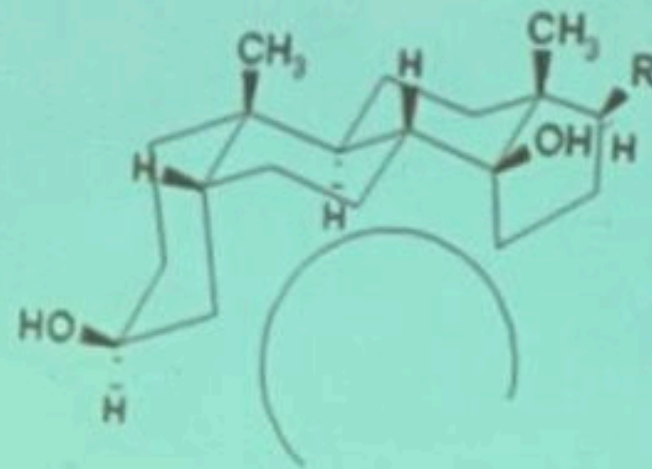
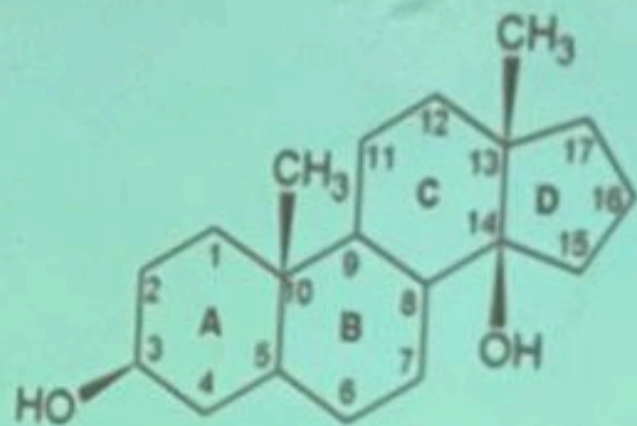
BUFADIENOLIDES (B)

bufalin $R = H$
proscillaridin A $R = Rha$



Structure Cont ...

- The unsaturated C-17 β -oriented lactone plays an important role in receptor binding
- Rings A/B and C/D are cis fused while rings B/C are trans fused.
- Such ring fusion gives the aglycone nucleus the **characteristic 'U' shape** as shown below which is **important for activity**



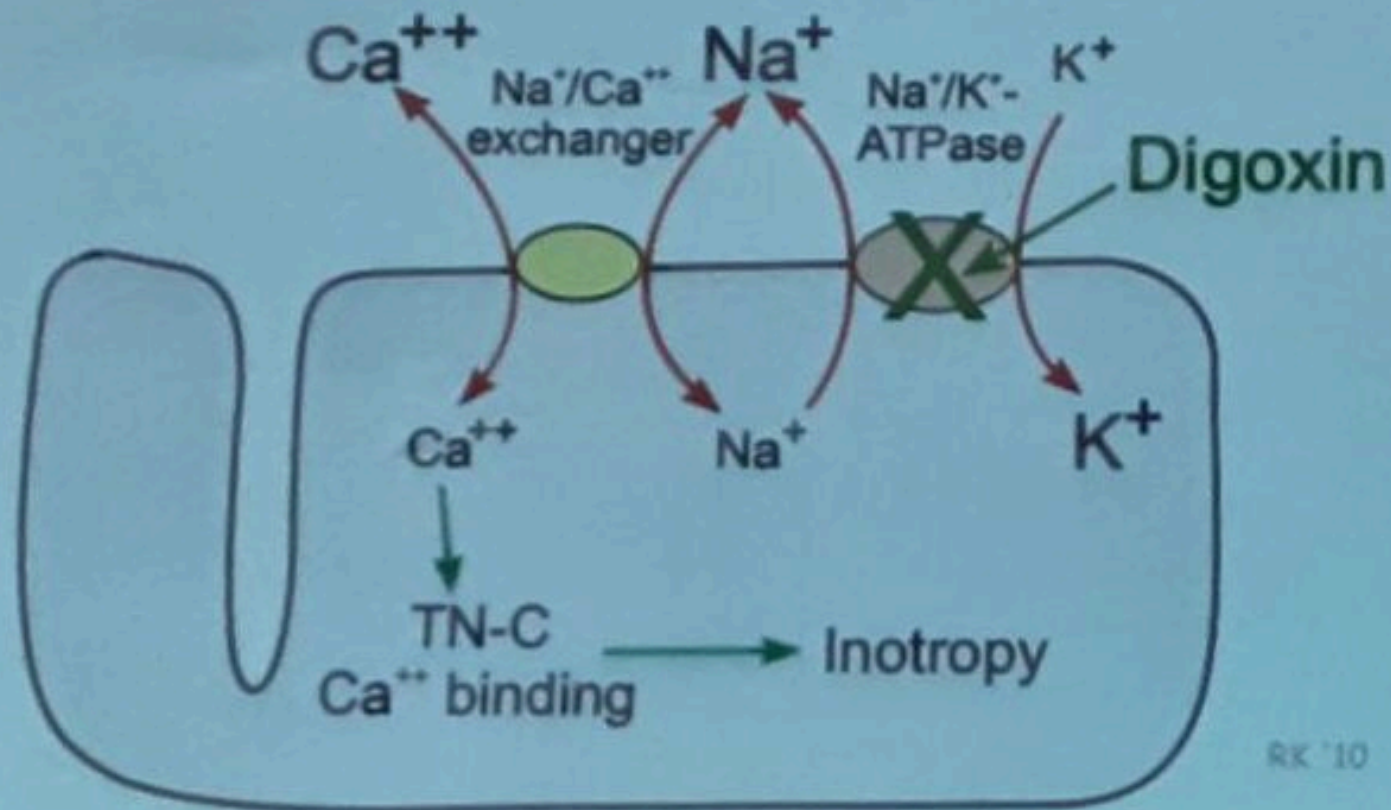
Sugars

- One to 4 sugars are found to be present in most cardiac glycosides attached to the 3β -OH group
- The sugars most commonly used include L-rhamnose, D-glucose, D-digitoxose, D-digitalose, D-digginose, D-sarmentose, L-vallarose, and D-fructose
- Free genins are absorbed faster and are metabolized more easily than their glycosylated counterparts Leading to **fast and short-lasting and (toxic effects)**
 - **Thus sugar part is important**

Function

- Cardiac glycosides work by **inhibiting the Na^+/K^+ pump**
- This causes an **increase in the level of sodium ions** in the myocytes, which then leads release of **Ca^{2+} from sarcoplasmic reticulum**
- This inhibition **increases the amount of Ca^{2+} ions available** for contraction of the heart muscle
 - improves cardiac output and reduces distention of the heart.

Biochemical Mechanism of Action



Sources

- Scrophulariaceae

- *Digitalis purpurea* leaves (foxglove)
- *Digitalis lanata* leaves – white flowers

- Apocyanaceae

- *Strophanthus* vine seeds – Africa

- Liliceae

- *Urginea* bulbs (squill) – Europe, India
- *Convallaria* leaves (lily of the valley) – also produces a volatile oil perfume



Digitalis

- Scrophulariaceae family
 - foxglove - biennial flowering plants
 - cases of poisoning rare
 - natural emetic if eaten in excess
 - *Digitalis purpurea* leaf – purple, British
 - -> Digitalis Tablets B.P.
 - -> Tincture of Digitalis B.P.
 - commercially grown Holland, E. Europe
 - NB no extraction for these products
 - *Digitalis lanata* leaf – white, Mediterranean
 - used for manufacture of pure glycosides
 - ie digoxin, lanatoside C
 - commercially produced Holland, Equador, USA



Detection

Color reactions due to aglycone moiety:

- These are characteristic for the **5-membered lactone ring** of cardenolides:
 - 1- **Kedde's test**: card. + Kedde's reagent (3,5-dinitrobenzoic acid + NaOH) → **violet colour**.
 - 2- **Baljet's test**: card. + Baljet's reagent (picric acid + NaOH) → **orange or red**.

Detection Cont...

- Reactions due to the steroidal nucleus:

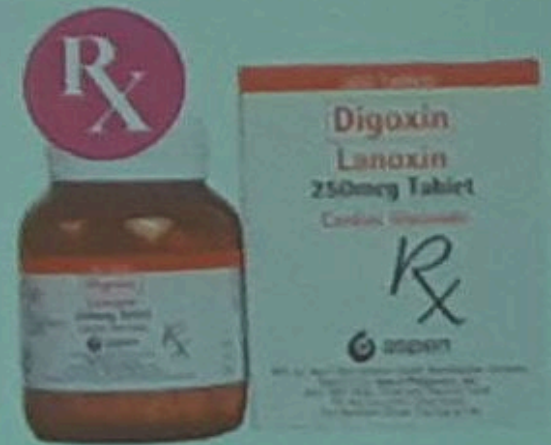
- **Liebermann's test:** card. in glacial acetic acid + $\text{H}_2\text{SO}_4 \rightarrow$ **red, violet, blue to green**

- Reactions due to the presence of **2-deoxy sugar** unit in the sugar moiety:

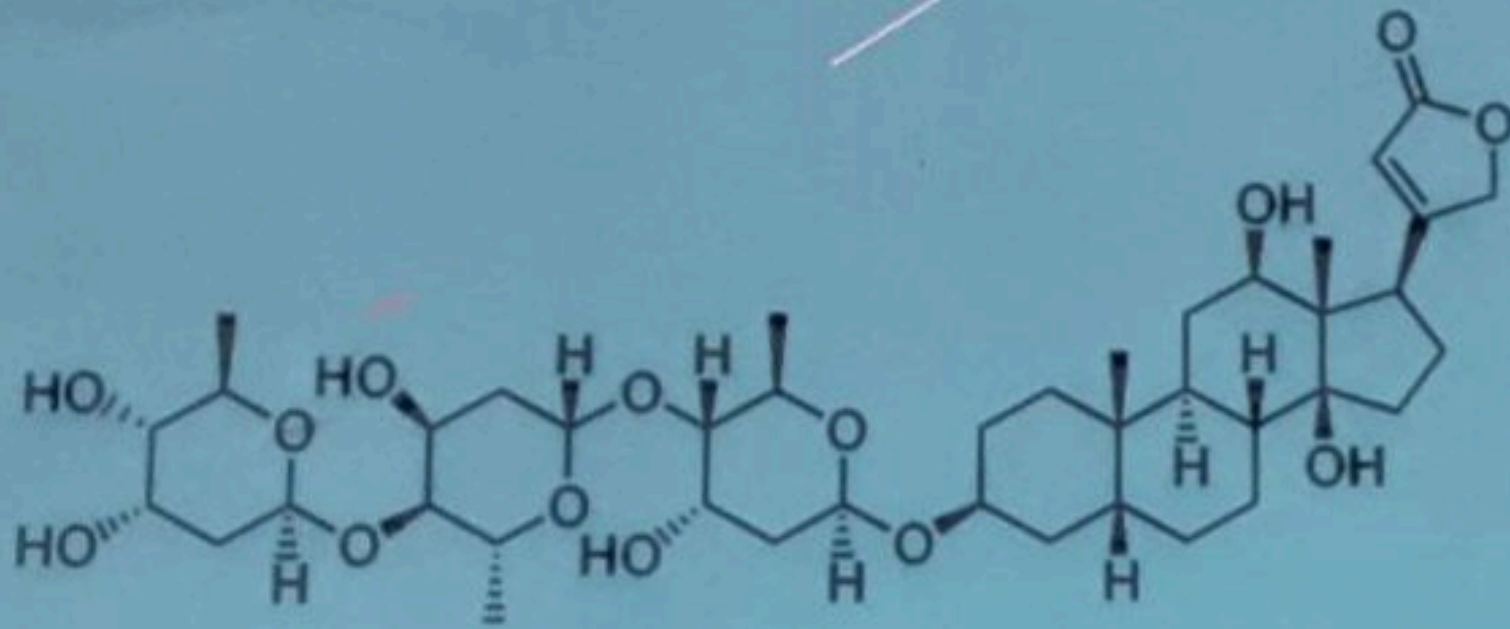
- **Keller-Killiani's test:** card. in glacial acetic acid containing traces of FeCl_3 , conc. H_2SO_4 on the wall of the test tube $\rightarrow$ **blue ring is formed in between the two layers**

Digoxin

- To treatment of various heart conditions,
 - **Atrial fibrillation, atrial flutter** and in **some cases heart failure**
- It is also available as a 0.05 mg/mL oral solution and 0.25 mg/mL or 0.5 mg/mL injectible solution and in tablet forms

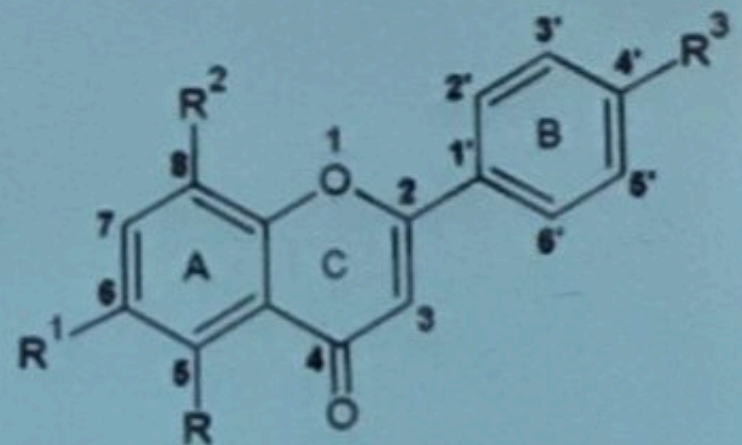


STRUCTURE

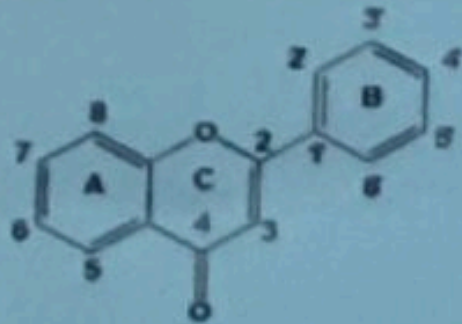


4. FLAVONOID GLYCOSIDES

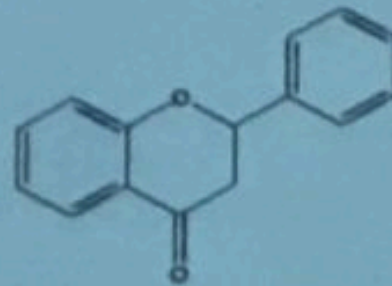
- The aglycone has a $C_6C_3-C_6$ nucleus
- Are usually substituted by **hydroxyl groups**
- Are responsible for the **color of flowers, fruits and leaves**
- Has a **2-phenyl-benzo- γ -pyrane (chromane)** nucleus consisting of **two benzene rings (A and B)** linked through a **heterocyclic pyran ring (C)**.
- Classified by the degree of
 - **hydroxylation, methoxylation, prenylation, glycosylation, and attachment of the B-ring in C-2 (most frequently) or C-3 position**



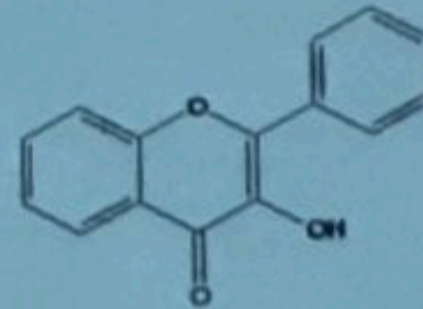
Structures



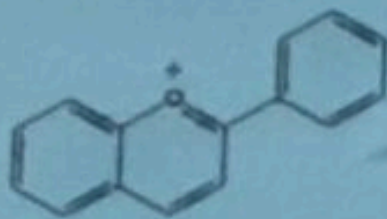
Flavone



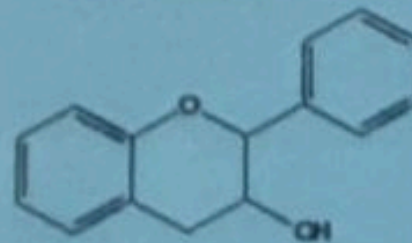
Flavanone



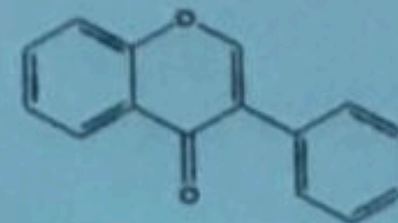
Flavonols



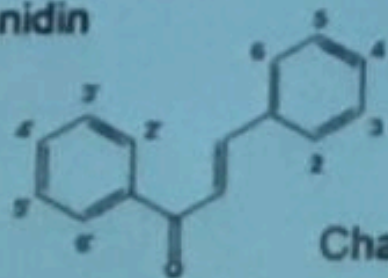
Anthocyanidin



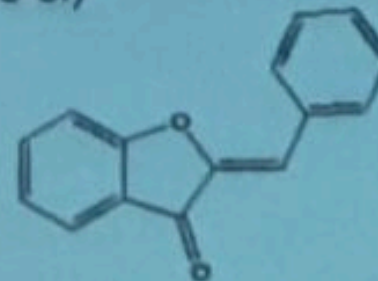
Catechin (flavan-3-ol)



Isoflavanoid



Chalcone



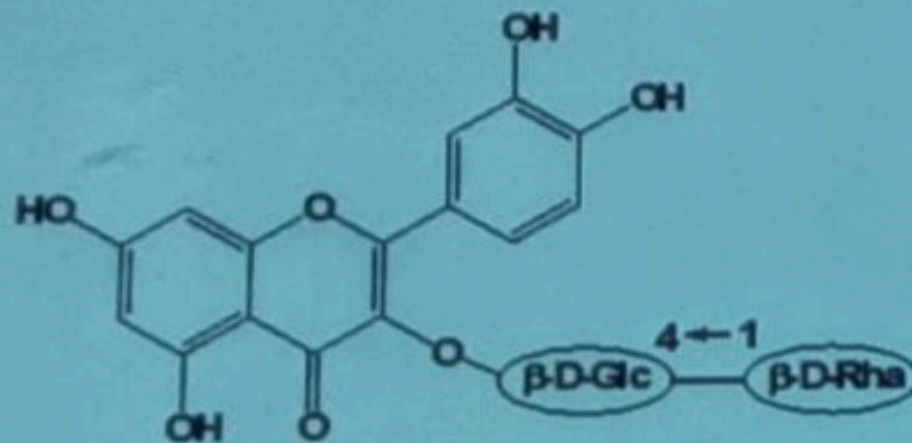
Aurones

- exhibit various **pharmacological activities including** anti-inflammatory (and analgesic) properties
 - anti-allergic effects
 - antithrombotic, vasoprotective properties
 - **decrease capillary fragility**
 - phlebitis – changes in vessel walls in extremities
 - > plasma leakage -> oedema
 - mainly due to high oestrogen, sometimes in males
 - tumour inhibition promotion
 - protective for gastric mucosa
 - antioxidant
 - anti-inflammatory, anti-HIV

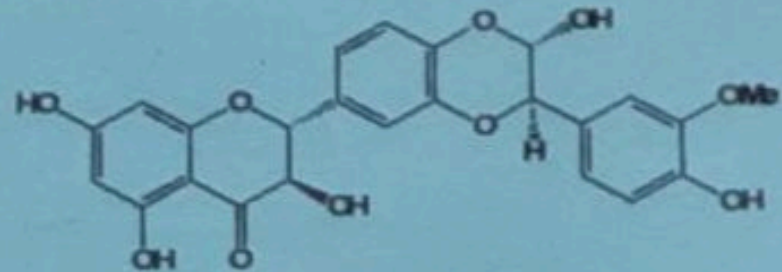
Some Medicinally Important Flavonoids

- **Rutin**

- Used as for the treatment of capillary bleeding and increased capillary fragility



- Silybin



- Silybin is the major active compound isolated from **silymarin**
 - a **hepato-protective** Flavonoidal fraction obtained from seed/fruits of *Silybum marianum* (Asteraceae) (Milk Thistle, St. Mary Thistle)
 - It is a mixture of mainly three flavonolignans, viz, silybin of, silidianin and silychristine.
- Silybin and also silymarin are used for wide range of liver problems

- Silymarin is included in the pharmacopoeia of many countries under the trademark LegalonTM or HepatronTM
- is often used as supportive therapy in food poisoning due to fungi and in chronic liver disorders, such as steatosis and alcohol-related liver disease

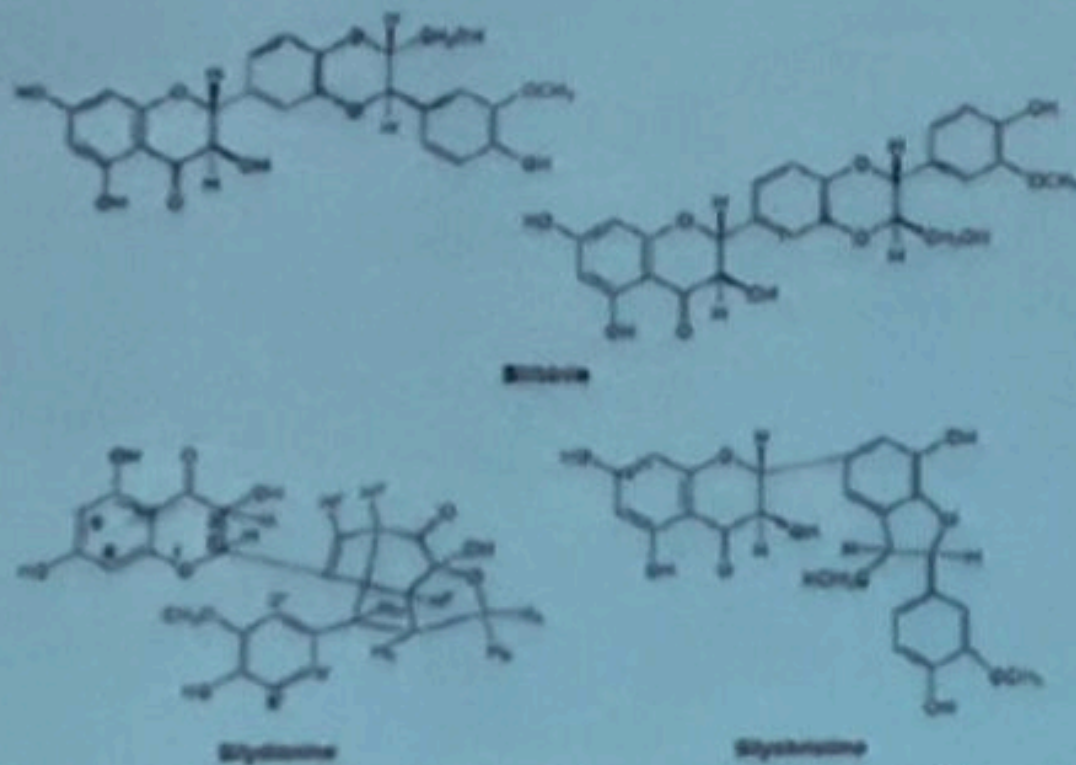
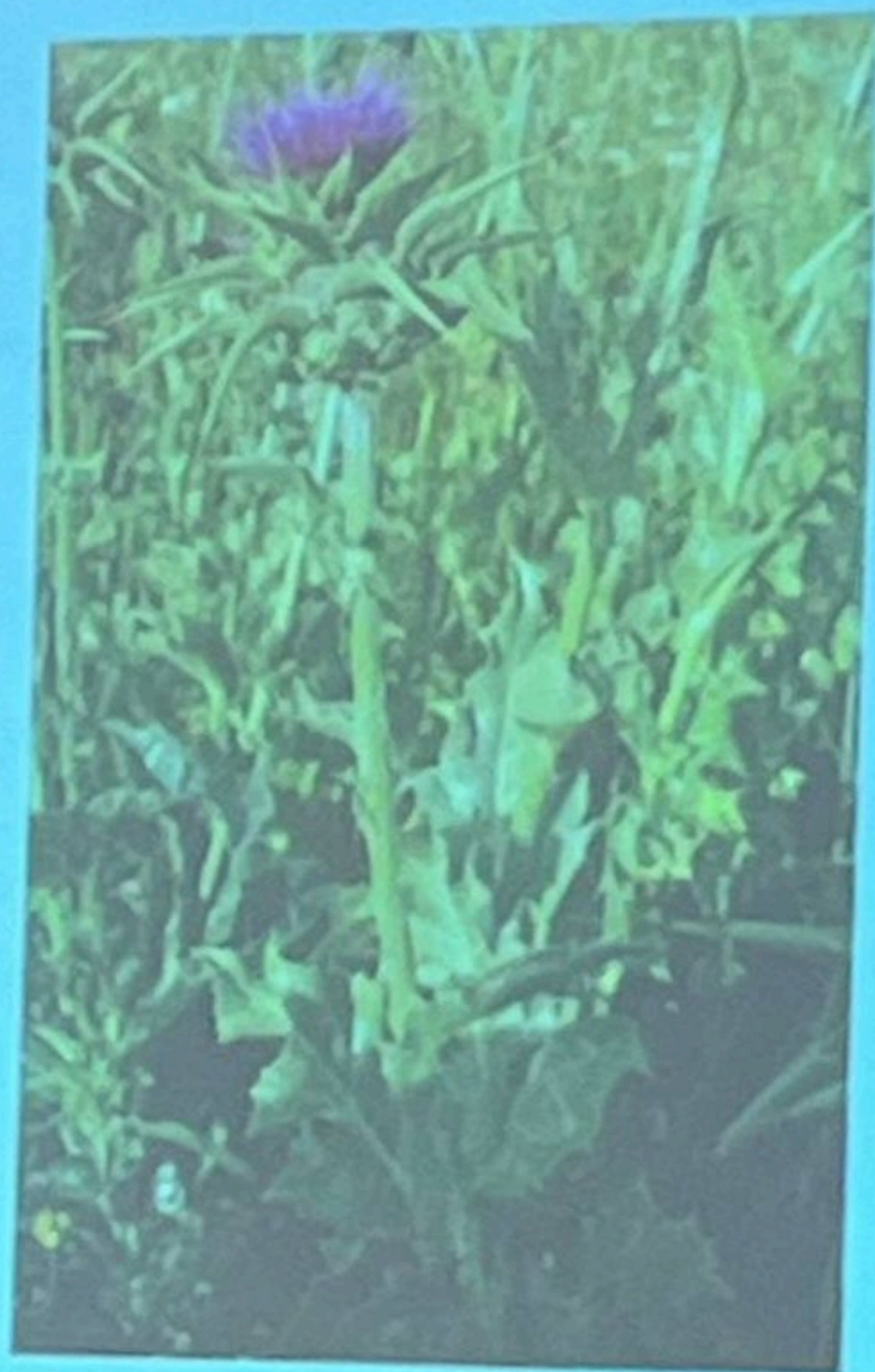


Figure 1: The three structural components of silymarin: silibinin, silydianine, and silychristine



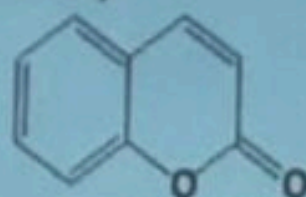
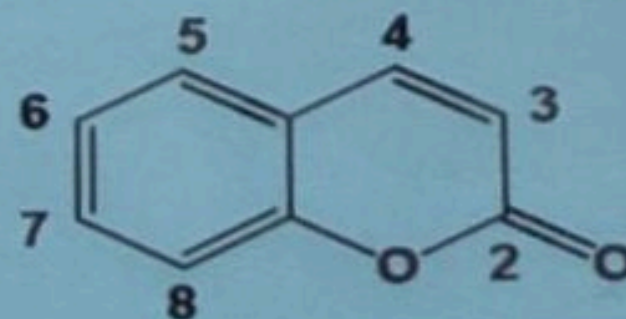
Isolation

- easy
- water and alcohol soluble
- give brightly colored solutions
- crystallize easily
- may give colour reactions eg
 - $\text{MgCl}_2 \rightarrow$ violet $\rightarrow$ orange
 - alkali KOH $\rightarrow$ orange
- easy to detect

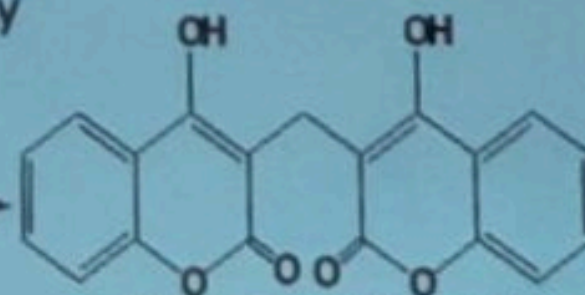


5. COUMARIN GLYCOSIDES (LACTONE GLYCOSIDES)

- some have anticoagulant activity
- They are widely distributed in nature
 - Leguminosae
 - Rubiaceae
 - Umbelliferae
 - Solanaceae
- They are converted to derivatives with such activity, by microbial action (e.g. dicoumarol)



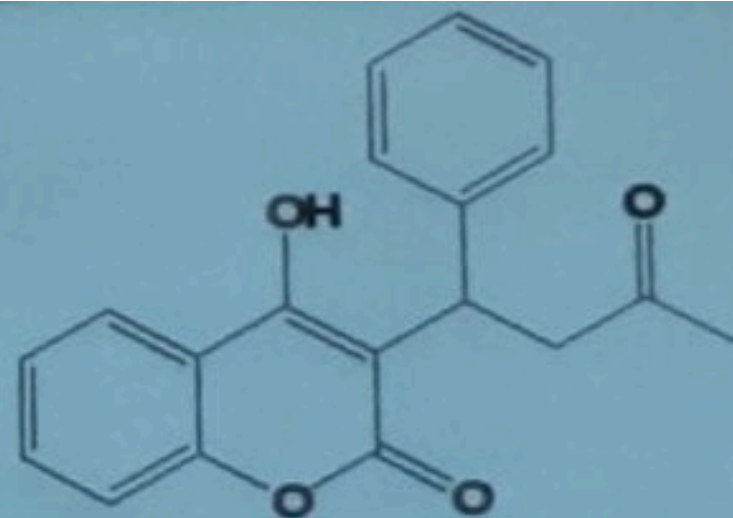
coumarin
does not have
anticoagulant activity



Dicoumarol (bishydroxycoumarin)
have strong
anticoagulant activity

- The structural requirements:

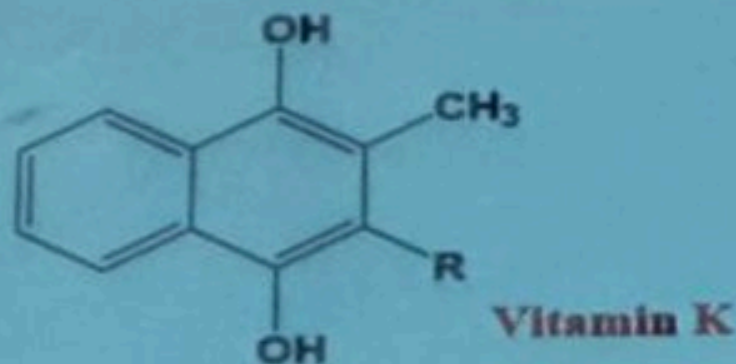
- **4-hydroxy group**
- **A substituent at C3.**



- Dicoumarol is very potent anticoagulant which can not be used for treatment purpose
- It was used as a lead compound for the synthesis of coumarin based oral anticoagulants such as **warfarin**

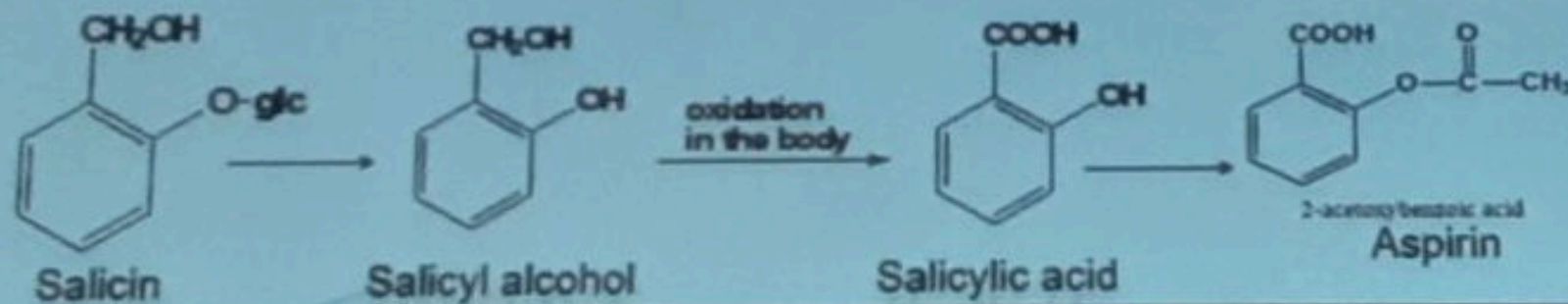
Mechanism of Action

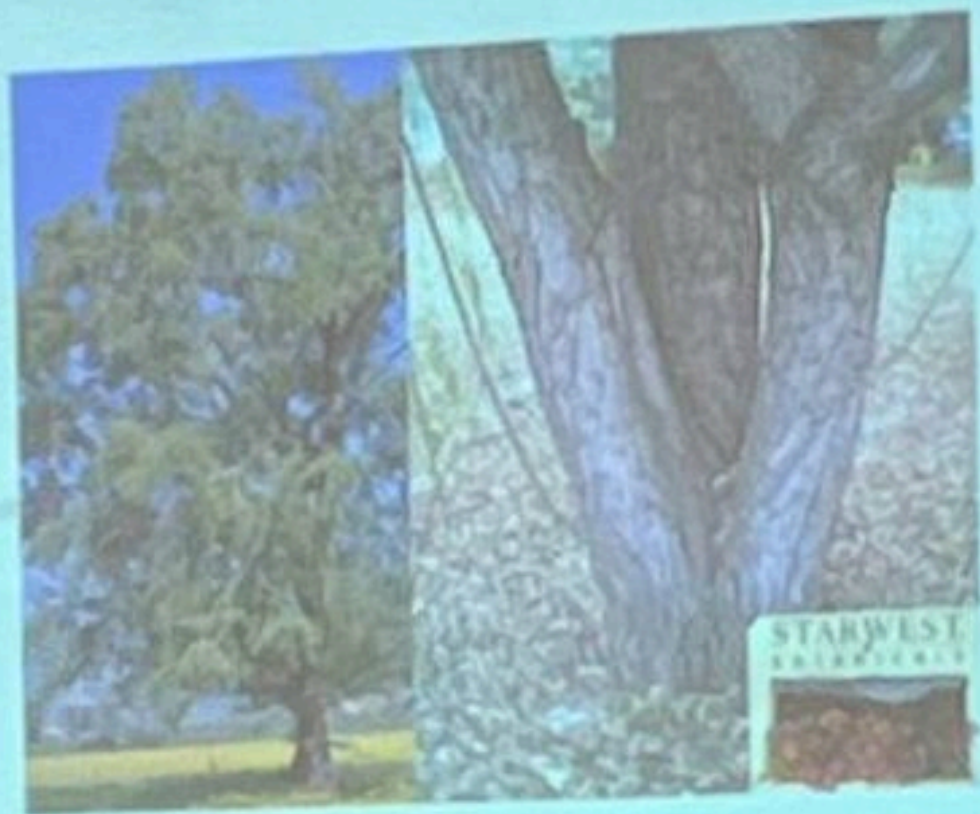
- are competitive inhibitors of vitamin K in the biosyntheses of functional thrombine.
- have structural resemblance to vitamin K



6. ALCOHOL GLYCOSIDES

- A notable example is **salicin**
 - isolated from **willow bark** (*Salix fragilis* or *S. purpurea*: fam. Salicaceae).
- **Traditionally willow bark** has been and still is used
 - Fever and flu-like symptoms
 - Analgesic (for headaches and toothaches)
 - Orally or topically for symptomatic relief of minor pains of the joint and rheumatoid disorders



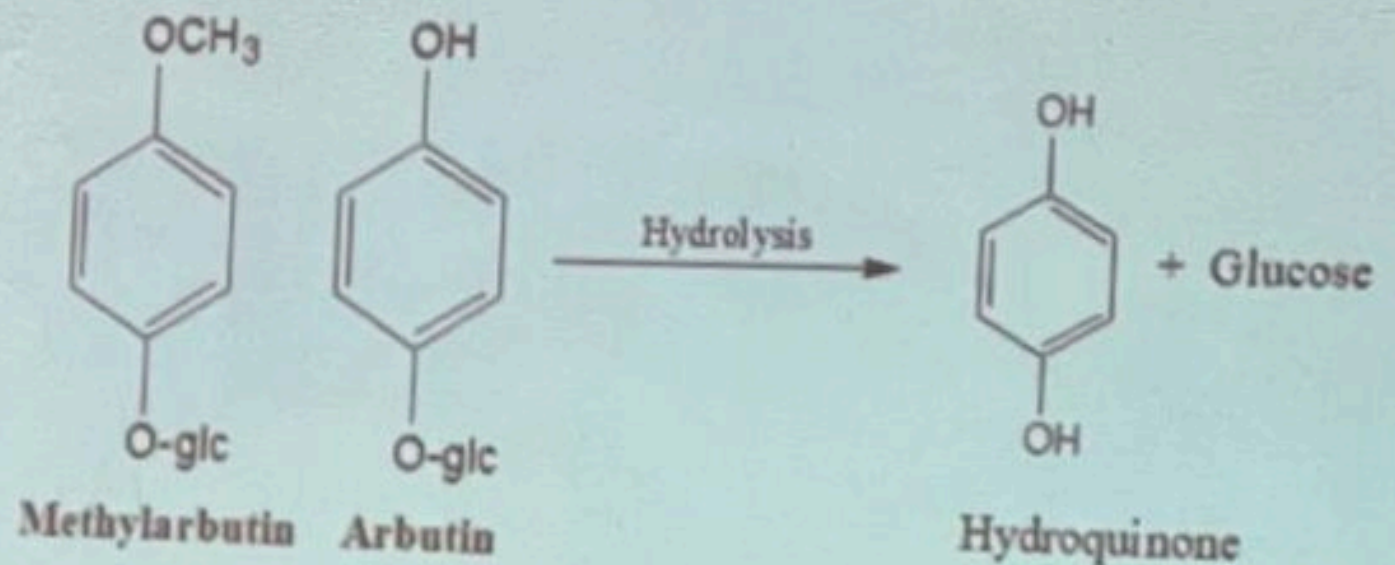


Arbutin & Methyl Arbutin

Source: *Uva ursi* (Bearberry leaves).

➤ **Nature:** Primary Phenolic glycoside (monoside)

➤ **Uses:** Diuretic- Bactericidal

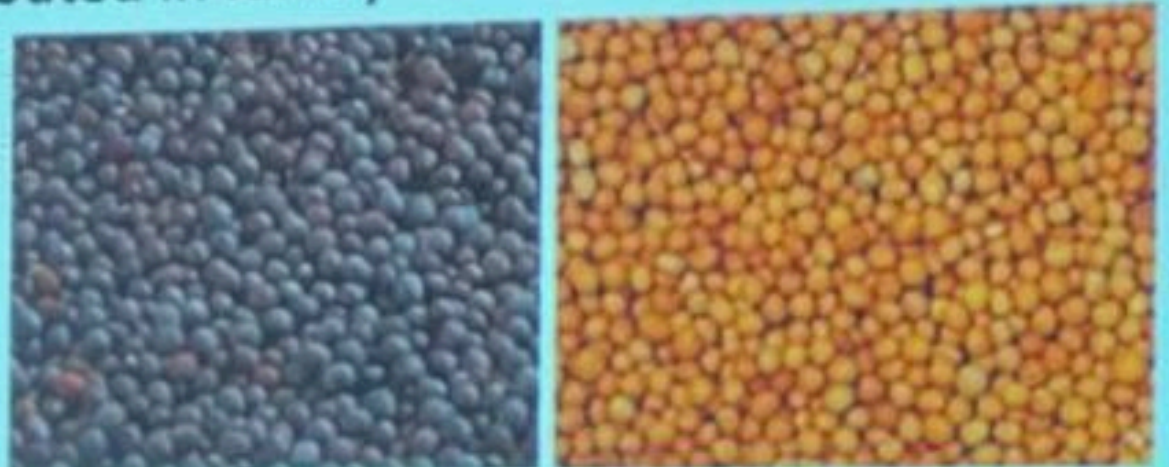
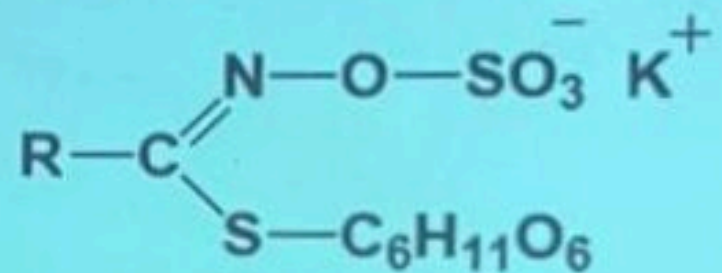




7. ISOTHIOCYANATE GLYCOSIDES (GLUCOSINOLATES OR THIOLGLUCOSIDES)

General formula:

- They are S-glycosides widely distributed in family Cruciferae
 - Can be separated on ion exchange resin
- 

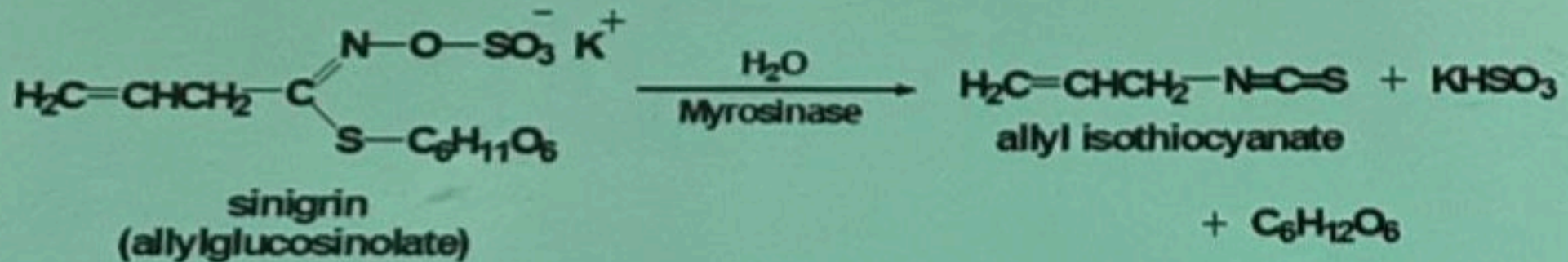


Sinigrin: In seeds of *Brassica nigra* (black mustard)

Sinabin: In Seeds of *Brassica alba* (white mustard)

Sinigrin

- Occurs in the seeds of black mustard (*Brassica nigra*: fam. Brassicaceae) or Indian mustard (*B. juncea*: fam. Brassicaceae)



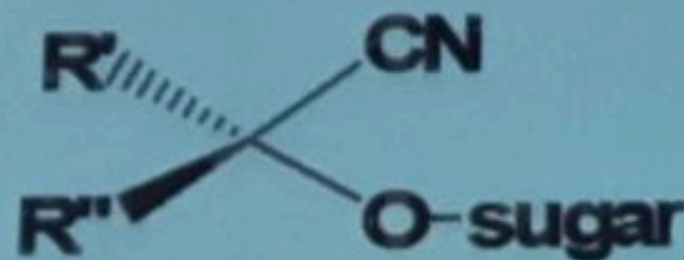
- Allyl isothiocyanate** is strong-smelling volatile compound
 - Responsible for the astringent odor and flavor of mustard

Uses

- Uses of allyl isothiocyanate
 - as condiment
 - is used as counter irritant (rubefacient)
 - used in the form of plasters

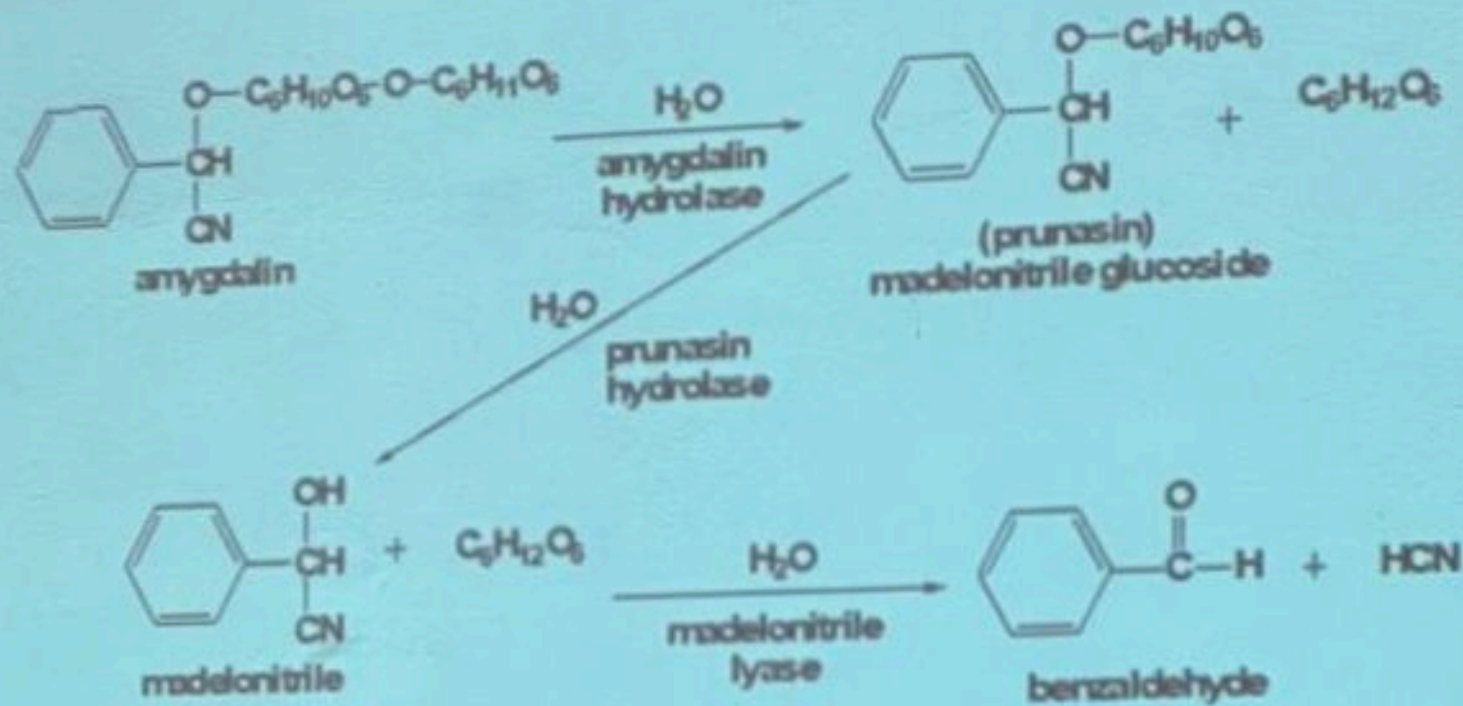
8. CYANOGENIC GLYCOSIDES

- Produce HCN upon complete hydrolysis which is toxic
 - Oral dose of 50 mg HCN can be fatal. How???
- All cyanogenic glycosides are of 2-hydroxynitriles



Occurrence and Distribution

- Over 2500 plant species
- Family Rosaceae, Fabaceae, Poaceae etc.
- Examples:
 - apricot , bitter almond, apple, plum, wild cherry (bark)
 - cassava (*Manihot esculenta*: Euphorbiaceae)
 - linseed (*Linum usitatissimum*: Linaceae)



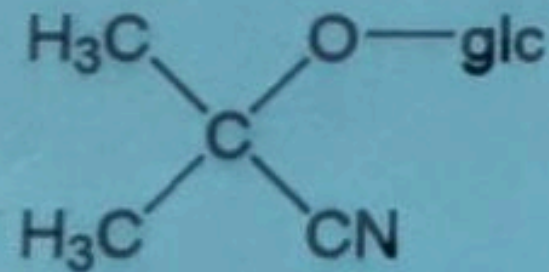
• Plants containing amygdaline are commonly employed for the preparation of benzaldehyde which is used as flavor

- **Cherry flavor**

Linamarin

Source: Linseed (Flaxseed)

➤ Structures: It is the glycosidic derivative of the cyanohydrin of acetone.



➤ Uses:

- Linamarin has a molluscicidal activity.
- Cyanogenic glycosides have role in cancer treatment



Test for Cyanogenic Glycosides

- Reduce plant material to small pieces and moisten with water
- Incubate at temp. less than 45 °C for 30 to 60 mins with the neck of the flask stoppered and have suspended sodium picrate paper
- The paper will turn brick red due to the release of HCN gas

9. ALDEHYDIC GLYCOSIDES

Glucovanillin

- Source: Vanilla pods.
- Uses: Flavouring agent- Spray reagent

